

Attaching Microlearning to Existing Habits

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ABSTRACT

To take advantage of microlearning, it is desirable for learners to make microlearning a habit. Most of the conventional methods for making microlearning a habit are intended to form a habit from scratch, but attempts to form a habit from scratch require a lot of motivation and workload from the learner. In this study, we propose habit-attached microlearning as a method to make microlearning a habit with less burden. This method designs a mechanism in which a microlearning task is added to the learner's existing habits. In this study, we focused on web browsing on smartphones as a habit to attach to learning, and developed a web browsing application for learning English vocabulary. This paper reports on a controlled experiment conducted to evaluate the learning and habit-forming effects of the developed application.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; **Empirical studies in HCI**; **Ubiquitous and mobile computing systems and tools**; • **Applied computing** → **Collaborative learning**; **E-learning**; *E-learning*.

KEYWORDS

Microlearning, Web browsing, Vocabulary building, Habituation

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1 INTRODUCTION

Microlearning has been attracting attention as a form of efficient learning. Microlearning is a form of learning in which learning content is divided into small portions and is applied using mobile devices such as smartphones [1]. One of the characteristics of mobile devices is that users can use them at their own convenience, regardless of their location. Therefore, the advantage of microlearning is that step-by-step learning can be achieved through the use of spare time. In order to take advantage of microlearning, it is

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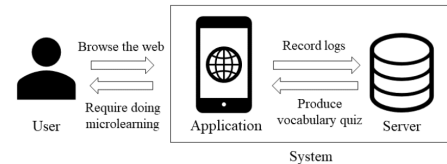


Figure 1: Overview of the proposed habit-attached microlearning system.

desirable for microlearning to become a habit, i.e., a state in which microlearning is continuously conducted.

Two methods have been devised to form learning habits using mobile devices. The first method is to use the notification function of the device. In this method, the system sends notifications to the user at the appropriate frequency and timing to encourage learning. The second method is gamification. This method incorporates game elements such as points and badges into the learning process. All of these methods are intended to form the habit of using a microlearning system from scratch. However, attempting to form learning habits from scratch requires a lot of motivation and effort from the user.

While it is difficult to form a habit, it has been suggested that once a habit is formed, it does not disappear easily [4]. Thus, it may be possible to form strong learning habits with less effort if learning can be incorporated into users' existing habits. In this study, we propose a new method called "habit-attached microlearning" as a way to make microlearning a habit. In this method, a mechanism is designed so that learning is added to the user's existing habits. Here, each learning task is assumed to be as small as possible. Habit-attached microlearning does not require much effort to form learning habits because it uses existing habits. In this study, we focus on web browsing as a habit to attach learning tasks. This is because many smartphone users frequently use web services such as SNS and video sharing sites. The purpose of this study is to develop a habit-attached microlearning system for web browsing.

In this paper, we propose a system that attaches an English vocabulary learning task to web browsing on a smartphone. We also report on an evaluation experiment of the proposed system. The research has been approved by the Ethics Committee of the Graduate School of Engineering, Osaka Prefecture University.

2 PROPOSED SYSTEM

In this section, we describe our proposed system. The proposed system is a mobile application with web browsing, word selection relevant to the web browsing behavior, and English word learning functions. Figure 1 shows an overview of the proposed system.

When a user browses the web using the proposed system, the system selects English words that are related to the web pages the user browses. The proposed system generates a learning task from the selected English words and asks the user to work on the learning task in between web browsing. The proposed system aims to increase the user's chances to work on the learning task by repeating this process. The aim of the proposed system is to increase the user's opportunities to work on learning tasks by repeating this process. The details of each function are described below.

2.1 Web Browsing Function

When the proposed system is launched, the home screen is displayed. On the home screen, the user can check bookmarks and browsing history. The user can display new tabs using bookmarks and search fields. In addition to browsing and transitioning to web pages, tabs allow users to register bookmarks and transition to the tab list screen. In the tab list screen, the user can move to the home screen and display or delete tabs. The proposed system has the same operability as a normal web browser, and can be used as a web browser without any sense of discomfort.

2.2 Relevant Word Selection Function

The proposed system selects English words relevant to the contents the user has viewed in the past as training words. The user can check the relevance of the selected English words to the viewed content when learning them. This feature is intended to enhance the learning effect and reduce the burden on the user by making the learning content follow the context of the user. Because we expect that original contents of the web are described in Japanese, WordNet is used to search for English translations of words and synonyms.

2.3 English Word Learning Function

The proposed system displays a question screen when a user moves from a tab to a tab list screen, or when the user moves a certain number of times from one web page to another. The question is a two-choice question in which the user answers the appropriate English word as a translation of the presented Japanese word. There are two types of questions. One is a question to collect unknown words selected by the proposed system. The other is a review question using words that have already been registered. When a user answers an unknown word collection question incorrectly, the word of the choice is registered as an unknown English word. The user can close the learning screen after answering a question correctly twice. The fact that the learning task is a simple two-choice question and the number of tasks is small is intended to make the learning task easier for the user to execute and to increase the continuity of use. In addition to the learning function by interrupting the user, the proposed system has a function to display a list of learned words and a function to ask a series of two-choice questions. This is to assist the user's independent review.

3 EXPERIMENT

This section describes the experiment conducted to evaluate the effectiveness of the proposed system. The purpose of the experiment was to verify whether the proposed system is effective for

learning and forming usage habits. Fourteen Japanese smartphone users participated in the experiment. Application A (App A) was distributed to 8 of the participants (4 males and 4 females; mean age: 23.9 ± 4.3 years). The remaining six participants (two males and four females; mean age: 22.5 ± 1.2 years) received Application B (App B). The participants used the application to collect and study unknown English words for 14 days. Participants were administered a test to measure the number of words they had learned and a questionnaire to assess their application usage habits every few days. The details of the experiment are described below.

3.1 Mobile Applications for the Experiment

This section describes the details of the distributed applications. App A is the proposed system; when asking questions to collect unknown words, we set up the system so that the questions include both words relevant to the user's browsing contents and words irrelevant to the contents, to evaluate the effect of relevant words. App B is the system built by removing the web browsing function from the proposed system and added a notification function. App B was set up so that users would receive a notification every 60 to 90 minutes to encourage them to learn English vocabulary. App B collected unknown words from a series of two-choice questions. The words selected by App B were independent of the user's behavior. The maximum number of English words that could be registered per day for both App A and App B was set to 20 words.

3.2 Experimental Procedure

This section describes the details of the experimental procedure. The experiment consisted of a 14-day learning period, including tests and questionnaires every few days. First, the application was distributed to the participants' smartphones, and the concept and usage of the application were explained orally while showing them the manual. During the explanation, the participants practiced using the application to improve their understanding. After explaining the application, we asked the participants to do the following three things during the study period.

- Collect 20 unknown English words using the application every day.
- Use the application to study during the study period.
- Answer a test and a questionnaire on days 4, 7, 10, and 13 of the study period and on the day after the study period (day 15 of the experiment).

Except for the above-mentioned requests, we did not ask anything including the timing of the use of the application. The tests and questionnaires were sent to the participants via email or chat in the morning of the days mentioned above. Participants were asked to complete the tests and questionnaires before using the applications on the same day. The test was created using the vocabulary collected by the participants by the day before the day of the experiment. The test consisted of writing the spelling of an English word from its Japanese translation having the first letter of the alphabet as a clue. The study period questionnaire measured the Self-Report Habit Index (SRHI) [3] in terms of application use. SRHI is a measure of the subjective strength of a habit. The questionnaire on the 15th day of the experiment asked about SRHI items as well as comfort with the application and feedback on the experiment.

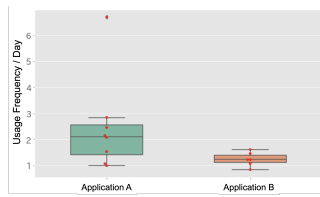


Figure 2: Frequency of usage of each application.

3.3 Experimental Results

Figure 2 shows the distribution of the number of times each application was launched per day from the second to the 14th day of the study. Application activation was defined as the case where a participant who had not used an application for more than 30 minutes started to operate it. The vertical axis of Figure 2 represents the number of application launches per day, and the red dots correspond to the number of launches for each participant. The box plot shows the distribution of the number of activations for all users of each application. As shown in this figure, the group that used App A tended to start the application more frequently than the group of App B. This suggests that attaching a learning task to web browsing may increase the opportunity for users to engage in learning more than prompting them to learn using notifications.

Next, we calculated the automaticity score of the participants' learning using the distributed application on the seventh day of learning. The automaticity score is the sum of the scores of the items that indicate the strength of automaticity of behavior in the SRHI [2]. The automaticity scores ranged from 0 to 42, and behaviors with an automaticity score of 21 or less could be considered non-habitual. The distribution of the automatism score on the 15th day showed that the use of both App A and App B was hardly habit-forming, indicating that there was no significant difference between the groups that used App A and App B during the experiment period.

Finally, we discuss the habit-forming effect of the proposed system based on the results of the questionnaire. One participant who used App A did not launch the application frequently during the learning period, but seemed to have been habitually using the web service with a smartphone before the experiment. This case suggests that some participants may not have been able to successfully attach the learning to their existing habit of web browsing because the proposed system was not an application they were familiar with. On the other hand, some participants tended to use App A as a web browser frequently. These participants responded in the questionnaire after the study period that they would like to continue using App A in the future, suggesting that their behavior was a sign that learning with the proposed system would become a habit. Based on the above considerations, it can be concluded that the attachment of the learning task by the proposed system may not necessarily be effective for all users, but for some users, long-term use of the proposed system can be expected to make learning a habit.

4 CONCLUSION

In this paper, we propose a system that attaches an English vocabulary learning task to web browsing on a smartphone. We evaluated the learning and habit-forming effects of the proposed system by

the experiment for 14 days. The experimental results suggest that attaching learning tasks to web browsing may increase users' opportunities to engage in learning, compared to prompting users to learn using notifications. In addition, it was suggested that some users can be expected to form learning habits after using the proposed system for a long time. Future work includes improving the method of attaching English vocabulary learning tasks and the UI in terms of improving the proposed system. From the viewpoint of the development of habit-attached microlearning, it is also necessary to devise a method of attaching learning tasks to habits other than web browsing.

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